

Translated

From: e-maxx.ru

Minimum stack / Minimum queue

In this article we will consider three problems: first we will modify a stack in a way that allows us to find the smallest element of the stack in $O(1)$, then we will do the same thing with a queue, and finally we will use these data structures to find the minimum in all subarrays of a fixed length in an array in $O(n)$

Stack modification

We want to modify the stack data structure in such a way, that it is possible to find the smallest element in the stack in $O(1)$ time, while maintaining the same asymptotic behavior for adding and removing elements from the stack. Quick reminder, on a stack we only add and remove elements on one end.

To do this, we will not only store the elements in the stack, but we will store them in pairs: the element itself and the minimum in the stack starting from this element and below.

```
stack<pair<int, int>> st;
```

It is clear that finding the minimum in the whole stack consists only of looking at the value `stack.top().second`.

It is also obvious that adding or removing a new element to the stack can be done in constant time.

Implementation:

- Adding an element:

```
int new_min = st.empty() ? new_elem : min(new_elem, st.top().second);  
st.push({new_elem, new_min});
```

- Removing an element:

```
int removed_element = st.top().first;  
st.pop();
```